Accessibility Info

What standard this pipeline targets, and why a clean checker is not the evidence

The standard, briefly

This pipeline targets **WCAG 2.1 Level AA**, the technical standard the DOJ's ADA Title II rule sets for web and mobile content at public universities (compliance date for larger institutions: April 26, 2027 — the requirements did not change with that date's extension, only the deadline did). **WCAG2ICT** is the clause that brings a standalone PDF into scope at all; without it, WCAG reads as a web-only standard. On the PDF side, that translates to **PDF/UA** conformance plus a set of Minimum Digital Accessibility Standards (MDAS) many universities check as a floor:

1. Accessibility permission flag is set (nothing blocks assistive tech).
2. The PDF has a title.
3. The PDF has full text — not image-only.
4. Document language is specified.
5. Every figure has alternative text.

If you are adapting this pipeline for a different institution, check its own accessibility policy for the specifics — the WCAG/PDF-UA/MDAS combination above is common but not universal, and `refs/` is where this kit keeps the source documents for whichever institution it targets, so an agent never has to guess or take a claim on faith.

UA-1 or UA-2 — this is a real choice, and it matters which

PDF/UA has two versions. **UA-2** (2024, requires PDF 2.0) has richer math tagging — equations can carry their MathML as an Associated File (`/AF`) rather than needing an explicit spoken alt text on every formula. **UA-1** (2014, PDF 1.4-1.7) predates that mechanism and does not recognize `/AF` as alternative text at all — a document declaring UA-2 with `/AF`-only math will show every equation without a hand-authored description as *failing* a UA-1 check.

This matters because **the checker your reviewer will actually run may only implement UA-1.** PAC 2026 — the tool many institutions point their reviewers at — checks against UA-1 specifically (confirmed by reading its own report header, not assumed). A document that is honestly, defensibly UA-2-conformant can still come back from an institutional review flagged for missing alternative text on nearly every equation, because the reviewer's tool was never checking what the document declared. **Match your declared standard to what will actually check it, not to which standard has the nicer feature set.** This kit's own `Multi_Agent_Outline.md` defaults to UA-1 for exactly this reason — see its Toolchain Prerequisites section for the full account of switching, including the kernel mechanism that makes it possible without hand-authoring hundreds of formula descriptions from scratch.

A conformance pass is a floor, not evidence

This is the single most important thing on this page, and it is not a platitude — it is a repeated, specific finding from building this pipeline:

- A checker returned a **clean pass three separate times** on a document that had 16 headings mis-tagged as ordinary paragraphs (no navigable outline at all), 24 content operators sitting completely outside the tag tree, and one table cell marked as decorative artifact — silently deleting the word "Quantum" from a Classical-vs-Quantum comparison and inverting its meaning for anyone using a screen reader.
- This pipeline's *own* verification script once reported "0 empty equations" while 8 equations were exporting nothing but a blank placeholder character, because the check compared *counts* instead of matching each formula to its own result by identity.
- A table's `/Summary` — required for an accessible data table — was written to the wrong place in the PDF's internal structure for months. Every internal test that checked "does a summary exist" said yes. The actual PDF specification does not recognize that location as valid at all, so a real accessibility reviewer's tool saw no summary whatsoever. Nothing about this was visible without checking the *specific location* a summary attribute is supposed to live in, not just whether the string was present somewhere in the file.

The pattern in all three: **the checker was answering a different, easier question than the one that mattered**, and every one of them returned a confident, clean, wrong answer. Treat any single tool's pass — including this kit's own scripts — as a floor to build from, never as proof of anything on its own.

What actually catches these things

`Multi_Agent_Outline.md`'s Step 4d runs veraPDF (the conformance floor) plus four mandatory numeric tests that are what actually caught every defect above:

1. **Structure-element inventory** — every heading in the source must produce exactly the matching number of tagged headings, correctly nested. (Catches: headings silently tagged as plain paragraphs.)
2. **Content reachability** — every piece of visible text must be either marked as decoration or reachable from the tag tree. Zero unreachable operators, no exceptions. (Catches: content technically present but invisible to assistive technology.)
3. **Artifact audit** — list everything marked as decorative and confirm none of it is actually meaningful content. (Catches: a data cell wrongly marked as page furniture.)
4. **Figure reading position** — no figure or table may cluster at the very end of the document's reading order regardless of where it sits visually on the page. (Catches: floats silently relocating their own tag-tree position away from the text that explains them.)

None of these are exotic. All four are scriptable (`tools/check_structure.py` runs them together) and none of them take more than a few seconds to run — the expensive part was ever discovering they were necessary, not running them once known.

When a checker itself is behind the standard

Institutional checkers update on their own schedule, not the standard's. If a checker's own dialog admits it does not fully support the PDF version your document uses, that is not something to argue with abstractly — verify directly whether the specific findings it reports still hold against your own file (structure-tree inspection, independent text extraction) before either dismissing them or chasing them blind. Sometimes the finding is real and the checker's version limitation is irrelevant to it; sometimes it isn't. The only way to know is to check, per finding, not to assume either way from the disclaimer alone.

Reviewer roles this pipeline uses, and why one of them cannot be overruled

`Multi_Agent_Outline.md` assigns an independent **Accessibility Auditor** whose APPROVE/REVISE verdict a physics content reviewer cannot waive, and the reverse: the auditor cannot waive a physics reviewer's verdict either. This split exists because visual quality and structural accessibility are genuinely independent

properties of the same file — a document can be typographically excellent and completely unusable with a screen reader, or vice versa — and a single reviewer optimizing for one will not reliably catch failures in the other. Expect the Accessibility Auditor to be the last reviewer still saying REVISE in most rounds. That is by design, not a bottleneck to route around.